Chapter 8

Differentiation

8.1 limits

The limit of a function $f(x)$ as x tends to (approaches) a number say "a" is defined as the value L of $f(x)$ as x approaches closer and closer to a without actually reaching it, and is denoted by

$$\lim_{x\to a} f(x) = L.$$

Example 8.1.1. Investigate the behavior of

$$f(x) = x^2,$$

as x approaches 2 from the left and from the right.

Solution

As we approach 2 from the negative(left), we obtain the following:

x	1.8	1.9	1.99	1.999	1.9999
$y = x^2$	3.24	3.61	3.9601	3.996001	3.99960001

From the table above, we observe that as x approaches 2 from the left, the values of the function $f(x) = x^2$ approach the value 4. Thus, we say that the limit as x approaches 2 from the left of $f(x)$ is 4, and write it as

$$\lim_{x\to 2^-} f(x) = \lim_{x\to 2^-} x^2 = 4$$

If we now approach 2 from the positive (right), we obtain the following:

x	2.2	2.1	2.01	2.001	2.0001
$y = x^2$	4.84	4.41	4.0401	4.004001	4.00040001

From the table above, we observe that as x approaches 2 from the right, the values of the function $f(x) = x^2$ approach the value 4. Thus, we say that the limit as x approaches 2 from the right of $f(x)$ is 4, and write it as

$$\lim_{x\to 2^+} f(x) = \lim_{x\to 2^+} x^2 = 4$$

Therefore, if $\lim_{x\to a^-} f(x) = L = \lim_{x\to a^+} f(x)$, then we say that the limit of $f(x)$ as x approaches a exists and is L. Thus

$$\lim_{x\to a} f(x) = L.$$

If $\lim_{x\to a^-} f(x) \neq \lim_{x\to a^+} f(x)$, then we say that the limit of $f(x)$ as x approaches a does not exist.

For "simple" functions such as the function considered in the previous example, we see that the value of the function at the point $x = a$ is the same as the value of the limit, and so

$$\lim_{x\to 2^-} x^2 = 4 = \lim_{x\to 2^+} x^2 = f(2) = (2)^2 = 4.$$

Example 8.1.2. Evaluate

1. $\lim_{x\to 2} 3x^2 + x - 5$

2. $\lim_{x\to 1} \sqrt{x}$

Solution

1. $\lim_{x\to 2} 3x^2 + x - 5 = 3(2)^2 + 2 - 5 = 9.$

2. $\lim_{x\to 1} \sqrt{x} = \sqrt{1} = 1.$

Some points to note:

1. We do not evaluate the limit by actually substituting $x = a$ in $f(x)$ in general, although in some simple cases it is possible.

2. The value of the limit can depend on which side it is approached, from left or right, i.e through the values of x less than a or through the values of x greater than a respectively. The two possible values may not be the same in which case the limit does not exist.

3. The limit may not exist at all and even if it does it may not be equal to $f(a)$.

Example 8.1.3. 1. If the function

$$f(x) := \begin{cases} x^2 - 1 & \text{if } x \leq 4 \\ x - 2 & \text{if } x > 4 \end{cases},$$

check whether the limit as x approaches 4 exist?

2. If the function

$$f(x) := \begin{cases} x^2 + 1 & \text{if } x \neq 2 \\ -4 & \text{if } x = 2 \end{cases},$$

(a) Find $f(2)$

(b) Find limit as $\lim_{x\to 2} f(x)$

Solutions

1. $\lim_{x\to 4^-} f(x) = \lim_{x\to 4}(x^2-1) = 4^2-1 = 15$. $\lim_{x\to 4^+} f(x) = \lim_{x\to 4}(x-2) = 4 - 2 = 2$. Since $\lim_{x\to 4^-} f(x) = 15 \neq \lim_{x\to 4^+} f(x) = 2$, $\lim_{x\to 4} f(x)$ does not exist.

2. (a) Clearly, $f(2) = -4$.

   (b) Since for all x different from 2, $f(x) = x^2 + 1$, $\lim_{x\to 2^-} f(x) = \lim_{x\to 2^+} f(x) = \lim_{x\to 2} x^2 + 1 = 2^2 + 1 = 5$.

Properties of limits

The properties of limits are fairly well what we might expect. Thus if $\lim_{x\to a} f(x) = b$ and $\lim_{x\to a} g(x) = c$, then

1. $\lim_{x\to a} kf(x) = k \lim_{x\to a} f(x) = kb$ for any constant k

2. $\lim_{x\to a}(f(x) \pm g(x)) = \lim_{x\to a} f(x) \pm \lim_{x\to a} g(x) = b \pm c$

3. $\lim_{x\to a}(f(x)g(x)) = \lim_{x\to a} f(x) \lim_{x\to a} g(x) = bc$

4. $\lim_{x\to a}(\frac{f(x)}{g(x)}) = \frac{b}{c}$, $c \neq 0$.

5. $\lim_{x\to a}(f(x))^n = (\lim_{x\to a} f(x))^n = b^n$.

Example 8.1.4. Evaluate each of the following.

1. $\lim_{x\to 2} 2x^3 + 4$.

2. $\lim_{x\to 2} \frac{x^2}{2x-1}$

3. $\lim_{x\to -7} \sqrt[3]{x-1}$

Solution

1. $\lim_{x\to 2} 2x^3 + 4 = \lim_{x\to 2} 2x^3 + \lim_{x\to 2} 4 = 2(2)^3 + 4 = 20$.

2. $\lim_{x\to 2} \frac{x^2}{2x-1} = \frac{\lim_{x\to 2} x^2}{\lim_{x\to 2} 2x-1} = \frac{2^2}{2(2)-1} = \frac{4}{3}$.

3. $\lim_{x\to -7} \sqrt[3]{x-1} = (\lim_{x\to -7} x - 1)^{\frac{1}{3}} = (-8)^{\frac{1}{3}} = -2$.

The kind of limits that cause most difficult and which are probably the most important are those arising from so called indeterminate forms. Any expression that yields results such as $0/0$, ∞/∞ or $0 \times \infty$ is called an indeterminate form. These include $\lim_{x\to 1} \frac{x^2-1}{x-1} = \frac{0}{0}$ and $\lim_{x\to 0} \frac{\sin x}{x} = \frac{0}{0}$.

Even though the function does not exist at such points, its limit at the point may exist. For example, $\lim_{x\to 1} \frac{x^2-1}{x-1} = 2$.

Example 8.1.5. Evaluate each of the following

1. $\lim_{x\to 1} \frac{x^2-1}{x-1}$

2. $\lim_{x\to 5} \frac{x^2-4x-5}{x-5}$.

Solution

1. Substituting the value of x gives $\frac{1^2-1}{1-1} = \frac{0}{0}$, which is an indeterminate form. Thus $\lim_{x\to1}\frac{x^2-1}{x-1} = \lim_{x\to1}\frac{(x-1)(x+1)}{x-1} = \lim_{x\to1}(x+1) = 1+1 = 2$.

2. Substituting the value of x gives $\frac{5^2-4(5)-5}{5-5} = \frac{0}{0}$, which is an indeterminate form. It follows that $\lim_{x\to5}\frac{x^2-4x-5}{x-5} = \lim_{x\to5}\frac{(x+1)(x-5)}{x-5} = \lim_{x\to5}(x+1) = 5+1 = 6$.

8.1.6 Infinite limits and limits at infinity

From the definition of limits, we observe the following:

1.
$$\lim_{x\to+\infty}\frac{1}{x} = 0$$

2.
$$\lim_{x\to0}\frac{1}{x} = \infty.$$

Example 8.1.7. Evaluate

1.
$$\lim_{x\to\infty}\frac{2x}{1+3x}.$$

2.
$$\lim_{x\to\infty}\frac{5-2x-9x^2}{2x^2+3}.$$

3.
$$\lim_{x\to\infty}\frac{\sqrt[3]{x^3+1}}{3x-5}.$$

Solutions

1.
$$\lim_{x\to\infty}\frac{2x}{1+3x} = \lim_{x\to\infty}\frac{x(2)}{x(\frac{1}{x}+3)} = \lim_{x\to\infty}\frac{2}{\frac{1}{x}+3} = \frac{2}{3}.$$

2.
$$\lim_{x\to\infty}\frac{5-2x-9x^2}{2x^2+3} = \lim_{x\to\infty}\frac{x^2(\frac{5}{x^2}-\frac{2}{x}-9)}{x^2(2+\frac{3}{x^2})} = \lim_{x\to\infty}\frac{\frac{5}{x^2}-\frac{2}{x}-9}{2+\frac{3}{x^2}} = -\frac{9}{2}.$$

3. HW

   Hint:
$$\lim_{x\to\infty} \frac{\sqrt[3]{x^3+1}}{3x-5} = \frac{1}{3}.$$

8.2 Continuity

Definition 8.2.1. A function $f(x)$ is said to be continuous at $x = a$ if and only if the following are satisfied

1. $f(a)$ is defined
2. $\lim_{x\to a} f(x)$ exist
3. $\lim_{x\to a} f(x) = f(a)$

Example 8.2.2. Investigate the continuity of each of the following:

1. $f(x) = \frac{x^2-1}{x+1}$ at $x = -1$
2. $$f(x) := \begin{cases} 2x+1 & \text{if } x \leq -1 \\ x^2-2 & \text{if } x > -1 \end{cases}, x = -1$$

Example 8.2.3. If the function
$$f(x) := \begin{cases} \frac{x^2-16}{x-4} & \text{if } x \neq 4 \\ C & \text{if } x = 4 \end{cases}$$
is continuous at $x = 4$, what is the value of C.

Example 8.2.4. Sketch the graph of
$$f(x) := \begin{cases} 2x+1 & \text{if } x \leq -1 \\ x^2-2 & \text{if } x > -1 \end{cases}.$$

Definition 8.3.1. Let f be a continuous function and a be a number in its domain. Then the derivative of f at a is the function $f'(x)$ defined by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h},$$

provided this limit exists.

8.3.2 First Principle

Given a function $y = f(x)$, the first principle states that the gradient function of $f(x)$ denoted by $f'(x)$ is given by

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}.$$

Example 8.3.3. Use the first principle to find $f'(x)$ in each of the following:

1. $f(x) = x + 1$
2. $f(x) = x^2$
3. $f(x) = \frac{1}{\sqrt{x}}$

Other notations in use include y' and $\frac{dy}{dx}$(read dee y by dee x) and is called the **derivative** or the **differential coefficient** of y with respect to x (abbreviated as wrt x). The process of finding the differential coefficient is called **differentiation**.

8.4 Differentiation of common functions

8.4.1 Polynomial functions

From differentiation by first principle of a number of examples such as the ones considered in the previous section, a general rule for differentiating $y = ax^n$ emerges, where a and n are constants. The rule is: if $y = ax^n$ then $\frac{dy}{dx} = anx^{n-1}$. By using this rule, it easy to check that the derivative of any constant is zero. In order to differentiate polynomials we recall that the derivative of a sum (difference) of functions is the sum (difference) of their derivatives. Thus, if $f(x) = p(x) + q(x) - r(x)$, (where f, p, q and r) are functions), then $f'(x) = p'(x) + q'(x) - r'(x)$.

Example 8.4.2. Find $f'(x)$ in each of the following:

1. $f(x) = x^2$
2. $f(x) = 3x^2 + 1$
3. $f(x) = \frac{1}{\sqrt{x}}$
4. $f(x) = 2 - 4x^2$

Example 8.4.3. Find the gradient of the function

$$f(x) = x^3 + 2x^2 + 1$$

at the point $(2, 3)$.

8.4.4 Trigonometric functions

Some important limits of trigonometric functions are

1. $$\lim_{x \to 0} \frac{\sin x}{x} = 1.$$

2. $$\lim_{x \to 0} \frac{\cos x - 1}{x} = 0.$$

By using the first principle, it can be shown that if $y = \sin x$ then $\frac{dy}{dx} = \cos x$, and if $y = \cos x$ then $\frac{dy}{dx} = -\sin x$. We will use these two differential coefficients to differentiate other trigonometric functions such as $\sec x$ and $\tan x$ among others. In general, if $y = \sin(f(x))$ then $\frac{dy}{dx} = f'(x)\cos(f(x))$, and if $y = \cos x$ then $\frac{dy}{dx} = -f'(x)\sin(f(x))$.

Example 8.4.5. Find $f'(x)$ in each of the following:

1. $f(x) = \sin x - \cos x$

2. $f(x) = \sin(x^2 + 2x - 4)$

3. $f(x) = \cos(\frac{1}{x})$.

8.4.6 Exponential Functions and Logarithmic Functions

It can be shown that if $y = e^{f(x)}$ and $y = \ln(f(x))$, then $y' = f'(x)e^{f(x)}$ and $y' = \frac{f'(x)}{f(x)}$, respectively. For a different base other than e, it can be shown that if $y = a^{f(x)}$ for some constant $a > 0$, then $y' = f'(x)a^{f(x)}\ln a$, and if $y = \log_b(f(x))$, then $y' = \frac{f'(x)}{(f(x))\ln b}$.

Example 8.4.7. Find $f'(x)$ in each of the following:

1. $y = e^x$

2. $y = e^{4x+5}$

3. $y = 6^{x^2}$

4. $y = \ln(x)$

5. $y = \ln(x^2 + 2x - 9)$

6. $y = \log_6(x^2 + 2x - 9)$

8.5 Methods of differentiation

In order to differentiate more complicated functions than the ones we have considered so far, such as products, quotients and composite functions, we will require the following results.

1. If y is a function of x, for example $y = f(g(x))$, it is often easier to make a substitution, $u = g(x)$ so that $y = f(u)$ before differentiating. It follows then that
$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}.$$
This is known as the 'function of a function' rule (or sometimes the **chain rule**).

Example 8.5.1. Differentiate the following:

(a) $y = 7\sin 2x$

(b) $y = (x^2 - 4x + 5)^8$

(c) $y = \sin^2(x)$

(d) $y = e^{t^3+2t^2-4t-9}$

(e) $y = \ln(2x + 1)$

Solutions

(a) We know that if $y = a\sin x$, then $\frac{dy}{dx} = a\cos x$. Thus, we apply the chain rule as follows: let $u = 2x$, so that $y = 7\sin u$. Thus $\frac{dy}{du} = 7\cos u$ and $\frac{du}{dx} = 2$. By

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx},$$

$$\frac{dy}{dx} = 7\cos 2x \cdot 2 = 14\cos 2x.$$

(b) $y = (x^2 - 4x + 5)^8$. Let $u = x^2 - 4x + 5$, so that $y = u^8$. Thus $u' = 2x - 4$ and $y' = 8u^7$. It follows that

$$\frac{dy}{dx} = 8(x^2 - 4x + 5)^7 \cdot (2x - 4)$$

by Chain rule.

(c) $y = \sin^2(x) = (\sin x)^2$. Let $u = \sin x$. Thus $y = u^2$. It follows that $\frac{dy}{du} = 2u$ and $\frac{du}{dx} = \cos x$. By

$$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx},$$

we get

$$\frac{dy}{dx} = 2\sin x\cos x.$$

(d) HW

(e) HW

2. When $y = uv$, and $u = f(x)$ and $v = g(x)$ are both functions of x, then
$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}.$$
This is known as the **product rule**.

Example 8.5.2. Differentiate the following:

(a) $y = (x^2 + 1)\sin^2 x$

(b) $y = e^{4x+5}\ln(x)$

(c) $y = x^2 \cos 3x$

Solutions

(a) $y = (x^2 + 1)\sin^2 x$. Let $u = x^2 + 1$ and $v = \sin^2 x$. We have that $\frac{du}{dx} = 2x$ and $\frac{dv}{dx} = 2\sin x \cos x$. By
$$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx},$$
we get
$$\frac{dy}{dx} = (x^2 + 1)\sin x \cos x + 2x\sin^2 x.$$

(b) $y = e^{4x+5}\ln(x)$. Let $u = e^{4x+5}$ and $v = \ln(x)$. From differentiation of exponential and logarithmic functions, we get $u' = 4e^{4x+5}$ and $v' = \frac{1}{x}$. Thus
$$\frac{dy}{dx} = \frac{e^{4x+5}}{x} + 4e^{4x+5}\ln(x).$$

(c) $y = x^2 \cos 3x$. Let $u = x^2$ and $v = \cos 3x$. We get $u' = 2x$ and $v' = -3\sin 3x$.. By Product rule,
$$\frac{dy}{dx} = -3x^2 \sin 3x + 2x\cos 3x..$$

3. When $y = \frac{u}{v}$, and $u = f(x)$ and $v = g(x)$ are both functions of x, then
$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}.$$

This is known as the **quotient rule.** Differentiate the following:

Example 8.5.3. (a) $y = \frac{3x^2}{2x-1}$.

(b) $y = \frac{e^x}{\sin x}$.

(c) $y = \frac{1}{x^3}$.

Solutions

(a) $y = \frac{3x^2}{2x-1}$. Let $u = 3x^2$ and $v = 2x - 1$. We get $\frac{du}{dx} = 6x$ and $\frac{dv}{dx} = 2$. By

$$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2},$$

$$\frac{dy}{dx} = \frac{6x(2x-1) - 6x^2}{(2x-1)^2} = \frac{6x^2 - 6x}{(2x-1)^2}.$$

(b) $y = \frac{e^x}{\sin x}$. Let $u = e^x$ and $v = \sin x$. We get $u' = e^x$ and $v' = \cos x$. Thus,

$$\frac{dy}{dx} = \frac{e^x \sin x - e^x \cos x}{\sin^2 x}.$$

(c) $y = \frac{1}{x^3}$. HW

When a function $y = f(x)$ is differentiated with respect to x, the differential coefficient is written as $\frac{dy}{dx}$ or $f'(x)$. If the expression is differentiated again, the second differential coefficient is obtained and is written as $\frac{d^2y}{dx^2}$ (pronounced dee two y by dee x squared) or $f''(x)$ (pronounced f double prime x).

8.5.4 Differentiation of implicit functions

The functions we have looked at so far were given explicitly; that is in the form $y = f(x)$ where the right hand side is only in terms of x. Implicit functions are functions such as $x^2 + xy^2 = \sin xy$. For such functions, it is often not easy to make y the subject of the formula for us to differentiate as before. It is possible to differentiate an implicit function by using the function of a function rule, which we looked at earlier.

Thus, to differentiate y^3 wrt x, the substitution $u = y^3$ is made, so that $\frac{du}{dy} = 3y^2$. Hence

$$\frac{d}{dx}(y^3) = \frac{du}{dy} \times \frac{dy}{dx} = 3y^2 \times \frac{dy}{dx}.$$

In summary,

$$\frac{d}{dx}(f(y)) = \frac{d}{dy}(f(y)) \times \frac{dy}{dx}.$$

When differentiating implicit functions containing products and quotients, the product and quotient rules of differentiation must be applied.

In the following example, we will learn how to differentiate implicit functions.

Example 8.5.5. 1. Find $\frac{dy}{dx}$ if $3x^2 + y^2 - 5x + y = 2$.

2. Given that $2y^2 - 5x^4 - 2 - 7y^3 = 0$, determine $\frac{dy}{dx}$.

3. If $4x^2 + 2xy^3 - 5y^2 = 0$, find $\frac{dy}{dx}$ at $(1, 2)$.

4. If $x \cos y + \frac{\sec y}{x^2} = \frac{5}{2}$, find $\frac{dy}{dx}$ at $(1, \frac{\pi}{3})$.

Solutions

1.

$$\begin{aligned}
\frac{d}{dx}(3x^2) + \frac{d}{dx}(y^2) - \frac{d}{dx}(5x) + \frac{d}{dx}(y) &= \frac{d}{dx}(2) \\
6x\frac{dx}{dx} + 2y\frac{dy}{dx} - 5\frac{dx}{dx} + \frac{dy}{dx} &= 0 \\
6x + 2y\frac{dy}{dx} - 5 + \frac{dy}{dx} &= 0 \\
2y\frac{dy}{dx} + \frac{dy}{dx} &= 5 - 6x \\
(2y + 1)\frac{dy}{dx} &= 5 - 6x \\
\frac{dy}{dx} &= \frac{5 - 6x}{2y + 1}.
\end{aligned}$$

2.

$$\begin{aligned}
2y^2 - 5x^4 - 2 - 7y^3 &= 0\\
4y\frac{dy}{dx} - 20x^3 - 21y^2\frac{dy}{dx} &= 0\\
4y\frac{dy}{dx} - 21y^2\frac{dy}{dx} &= 20x^3\\
\frac{dy}{dx} &= \frac{20x^3}{4y - 21y^2}.
\end{aligned}$$

3. Here, the second term involves a product, and so the product rule will be applied.

$$\begin{aligned}
4x^2 + 2xy^3 - 5y^2 &= 0\\
8x + 6xy^2\frac{dy}{dx} + 2y^3 - 10y\frac{dy}{dx} &= 0\\
8x + 2y^3 &= 10y\frac{dy}{dx} - 6xy^2\frac{dy}{dx}\\
10y\frac{dy}{dx} - 6xy^2\frac{dy}{dx} &= 8x + 2y^3\\
\frac{dy}{dx} &= \frac{8x + 2y^3}{10y - 6xy^2}\\
\frac{dy}{dx} &= \frac{4x + y^3}{5y - 3xy^2}.
\end{aligned}$$

Hence, at $(1, 2)$,

$$\frac{dy}{dx} = \frac{4(1) + 2^3}{5(2) - 3(1)2^2} = \frac{12}{10 - 12} = -6.$$

4. HW.

8.6 Some Applications of Differentiation

In this chapter, we present some applications of differentiation in our day to day life. These include finding the slope or gradient of a curve at a given point, sketching curves, rate of change, optimization problems, i.e. problems of determining the maximum or minimum point of a function.

8.6.1 The Gradient

We recall that the gradient function of a curve $y = f(x)$ at a point (a, b), denoted by $f'(a)$, is given by

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}.$$

Example 8.6.2. Find the gradient of the curve

$$y = \frac{1}{3}x^2 - 2x$$

at $(1, 2)$.

Solution If $y = \frac{1}{3}x^2 - 2x$, then the gradient function is given by $\frac{dy}{dx} = \frac{2}{3}x - 2$. Thus, at $(1, 2)$, the gradient is $m = -\frac{4}{3}$.

8.7 Tangents and Normals

8.7.1 Tangents

The equation of the tangent to the curve $y = f(x)$ at the point (x_1, y_1) is given by:

$$(y - y_1) = m(x - x_1),$$

where $m = \frac{dy}{dx}$ =gradient of the curve at (x_1, y_1).

Example 8.7.2. Find the equation of the tangent to the curve $y = x^2 - x - 2$ at the point $(1, -2)$.

Solution

$$(y - y_1) = m(x - x_1),$$

where $m = \frac{dy}{dx_{x=1}} = 2(1) - 1 = 1$. Thus

$$y - (-2) = 1(x - 1)$$
$$y = x - 3.$$

8.7.3 Normals

The normal at any point on a curve is the line which passes through the point and is at right angles to the tangent. It may be shown that if two lines are at right angles, then the product of their gradients is -1. Thus if m is the gradient of the tangent, then the gradient of the normal is $\frac{-1}{m}$. Hence the equation of the normal at the point (x_1, y_1) is given by:

$$(y - y_1) = \frac{-1}{m}(x - x_1).$$

Example 8.7.4. Find the equation of the normal to the curve $y = x^2 - x - 2$ at the point $(1, -2)$.

Solutions $\frac{dy}{dx} = 2x - 1$, so that $m = 1$.

$$\begin{aligned} y - (-2) &= \frac{-1}{1}(x - 1) \\ y &= -x - 1. \end{aligned}$$

Example 8.7.5. Determine the equations of the tangent and normal to the curve $y = \frac{x^3}{5}$ at the point $(-1, \frac{-1}{5})$.

Solution HW.

8.8 Rates of change (Related Rates)

If a quantity y depends on and varies with a quantity x, then the rate of y with respect to x is $\frac{dy}{dx}$. Thus, for example, the rate of change of pressure P with height h is $\frac{dP}{dh}$. If the rate of change is with respect to time, we usually call it 'the rate of change', the 'with respect to time', being assumed. Therefore, the rate of change of current, i, is $\frac{di}{dt}$ and the rate of change of temperature, θ, is $\frac{d\theta}{dt}$, and so on.

Example 8.8.1. The length l meters of a certain metal rod at temperature $\theta°C$ is given by $l = 1 + 0.00005\theta + 0.0000004\theta^2$. Determine the rate of change of length, in $mm/°C$, when the temperature is

1. $100°C$

2. $400°C$.

Solution

$l = 1 + 0.000050\theta + 0.0000004\theta^2$. $\frac{dl}{d\theta} = 0.00005 + 0.0000008\theta$.

1. When $\theta = 100°$, $\frac{dl}{d\theta} = 0.00005 + 0.00008 = 0.00013$.

2. When $\theta = 400°$, $\frac{dl}{d\theta} = 0.00005 + 0.00032 = 0.00037$.

Example 8.8.2. The luminous intensity I candelas of a lamp at varying voltage V is given by $I = 4 \times 10^{-4}V^2$. Determine the voltage at which the light is increasing at a rate of 0.6 candelas per volt.

Solution

HW

8.9 Velocity and Acceleration

When a car moves a distance x meters in a time t seconds along a straight road, if the velocity is constant, then $V = \frac{x}{t}$ m/s, i.e the gradient of the distance/time graph is constant. If, however, the velocity of the car is not constant, then the distance/time graph will not be a straight line. Hence, the velocity of the car at any instant is given by the gradient of the distance time graph. If an expression for the distance x is known in terms of time t, then the velocity is obtained by differentiating the expression. The acceleration a of the car is defined as the rate of change of velocity. Thus, if an expression for velocity is known in terms of time t, then the acceleration is obtained by differentiating the expression.

Example 8.9.1. The distance x meters moved by a car in a time t seconds is given by $x = 3t^3 - 4t - 1$. Determine the velocity and acceleration when

1. $t = 0$

2. $t = 1.5\,s$.

Solutions Given the distance function, $x = 3t^3 - 4t - 1$, we define the velocity, v, and acceleration, a, as follows:

$$v: \frac{dx}{dt} = 9t^2 - 4 \text{ and}$$
$$a: \frac{dv}{dt} = 18t.$$

Consequently,

1. at $t = 0$, $v = -4\,m/s$ and $a = 0\,m\ s^2$.
2. at $t = 1.5$, $v = \frac{65}{4}\,m/s$ and $a = 27\,m\ s^2$.

Example 8.9.2. Supplies are dropped from a helicopter and the distance fallen in a time t seconds is given by $x = \frac{1}{2}gt^2$, where $g = 9.8m/s^2$. Determine the velocity and acceleration of the supplies after it has fallen for 2 seconds.

Solution

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Example 8.9.3. The distance x meters traveled by a vehicle in time t seconds after the brakes are applied is given by $x = 20t - \frac{5}{3}t^2$. Determine

1. the speed of the vehicle (in Km/h) at the instant the brakes are applied,
2. the distance the car travels before it stops.

Solution

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8.9.4 Increasing and Decreasing Functions

Let $y = f(x)$ be a function continuous on a given interval. We have that

1. If
$$\frac{dy}{dx} > 0,$$
then $y = f(x)$ is increasing in that interval.

2. If
$$\frac{dy}{dx} < 0,$$
then $y = f(x)$ is decreasing in that interval.

Critical values are the values of x for which $f'(x) = 0$, at a given point.

Example 8.9.5. Given that

$$y = 2x^3 - 3x^2 - 12x,$$

1. Find the critical values of $f(x)$.
2. Determine the intervals where $f(x)$ is decreasing and where it is increasing.
3. Hence sketch the graph of $f(x)$.

Solution

1. If $y = 2x^3 - 3x^2 - 12x$, then $\frac{dy}{dx} = 6x^2 - 6x - 12$. From $6x^2 - 6x - 12 = 0$, we get $x = -1, x = 2$.
2. The critical values divide the number line into three intervals $(-\infty, -1)$ and $(2, \infty)$. By choosing a testing value, x value from each interval, it is easy to show that $y = 2x^3 - 3x^2 - 12x$ is increasing in $(-\infty, -1)$, and $(2, \infty)$, and decreasing in $(-1, 2)$.
3. It is easy to show that $2x^3 - 3x^2 - 12x = 0$ implies that
$$x = 0,\ x = \frac{3 \pm \sqrt{105}}{4},$$
as the x intercepts. From the study of polynomial functions, the sketch then follows.

8.9.6 Stationary Points

The gradient (or rate of change) of the curve may be positive or negative at a given interval or seen to be zero at a point, say P. Thus, if at

P the gradient is zero, and, as x increases, the gradient of the curve changes from positive just before P to negative just after, then P is called the **maximum point**, and appears as the 'crest of a wave'. On the other hand, if at Q the gradient is zero, and, as x increases, the gradient of the curve changes from negative just before Q to positive just after, then Q is called the **minimum point**, and appears as the 'bottom of a valley'. Points such as P and Q are given a general name of **Turning points.**

It is possible to have a turning point, the gradient on either side of which is the same. Such a turning point is given the special name inflexion point. Maximum and minimum points and points of inflection/inflexion are given the general term of **stationary points.**

Procedure for finding and distinguishing between stationary points.

1. Given a function $y = f(x)$, find $\frac{dy}{dx}$.
2. Let $\frac{dy}{dx} = 0$ and solve for x.
3. Substitute the value(s) of x found in (2), in the original equation $y = f(x)$, to get the corresponding y value(s).
4. To determine the nature of the stationary points. Either
   (a) **Find** $\frac{d^2y}{dx^2}$ and substitute into it the value(s) of x found in (2). If the result is:
      i. positive, the point is a minimum on,
      ii. negative, then the point is a maximum one,
      iii. zero, then the point is a point of inflexion.

   (b) **Use the sign of the gradient** of the curve just before and just after the stationary point.

Example 8.9.7. Locate the turning point on the curve $y = 3x^2 - 6x$ and determine its nature by examining the sign on either side.

Solution From $y = 3x^2 - 6x$, we get $\frac{dy}{dx} = 2x - 6$, so that $x = 3$ is the only critical value. Thus $(3, 9)$ is the only turning point.

To determine the nature of this turning point, we observe that from $\frac{dy}{dx} = 2x - 6$, $\frac{d^2y}{dx^2} = 2$, which is positive at that point. Thus, $(3, 9)$ is a minimum turning point.

Example 8.9.8. Locate the turning point of the following curve and determine whether it is maximum or minimum point, $y = 4\theta + e^{-\theta}$.

Solution

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Example 8.9.9. Find the maximum and minimum values of the curve $y = x^3 - 3x + 5$.

Example 8.9.10. Determine the coordinates of the maximum and minimum points of the graph $y = \frac{x^3}{3} - \frac{x^2}{2} - 6x + \frac{5}{3}$ and distinguish them. Sketch the graph.

Solution

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NOTE: THESE LECTURE NOTES ON DIFFERENTIAL CALCULUS ARE MEANT FOR MAT1110 (2020/2021) ACADEMIC YEAR ONLY.